

Revisiting the Intrusion Detection in In-Vehicle Networks

Muhammad Asif Khan¹, Senior Member, IEEE, Hamid Menouar, Senior Member, IEEE, and Mohamed Abdallah², Senior Member, IEEE

Abstract—An in-vehicle network (IVN) is the internal communication network that connects all sensors and control units in an autonomous vehicle. Sensors and control units use the IVN to send perception-related messages and control commands for the normal and safe operation of the vehicle. However, the IVN, by design, is vulnerable to network attacks due to a lack of adequate security mechanisms. This paper presents a Dynamic Windowing Intrusion Detection System (DWIDS) that adapts its detection window in real-time based on observed anomalies, enabling accurate and responsive attack detection. Unlike prior methods that focus on static configurations or single-attack detection, DWIDS supports multi-label classification and real-time tuning of detection parameters. The system is evaluated using two public benchmark datasets (CHD and IVN-IDS challenge) which feature diverse and imbalanced attack types. Experimental results demonstrate high performance across key metrics (e.g., >98% precision, >97% recall and F1-score), including for rare attacks. The findings confirm DWIDS's practicality and robustness for deployment in real-world autonomous vehicle environments.

Index Terms—Attacks, autonomous vehicles, CAN bus, ECU, in-vehicle network, IVN, intrusion detection.

I. INTRODUCTION

AUTONOMOUS vehicles (AVs) are becoming commonplace in today's transportation system, revolutionizing the way people and goods move. These vehicles leverage advanced technologies such as artificial intelligence, machine learning, computer vision, and sensor fusion to perceive their environment and make real-time decisions without human intervention. AVs promise to enhance road safety by minimizing accidents caused by human error, improve traffic efficiency through smart routing, and reduce carbon emissions by promoting eco-friendly driving patterns. However, the widespread adoption of AVs also presents challenges, including cybersecurity concerns, particularly related to in-vehicle networks (IVNs) [1], [2], [3]. Intrusion detection in IVNs is a critical safety requirement, yet existing solutions often rely on unrealistic assumptions or lack scalability for

real-time, multi-attack environments. This study addresses these limitations by proposing a practical, generalizable, and adaptive IDS design tailored for deployment in real-world autonomous vehicle platforms.

An IVN connects various electronic control units (ECUs), sensors, and actuators, facilitating real-time communication for autonomous driving functions.

The Controller Area Network (CAN) [4] is a communication protocol for in-vehicle networks developed by Bosch in the 1980s. CAN allows the various sensors and electronic control units (ECUs) in a vehicle to communicate with each other via a single two-wire system. CAN is also used in industrial automation, medical equipment, and other fields.

The CAN bus protocol was not designed with security in mind and thus has some intrinsic attributes that make it vulnerable to attacks. Although alternative communication protocols have been developed to address its shortcomings, CAN-FD (Flexible Data-rate CAN) [3] offers higher bandwidth and supports longer payloads, allowing for the inclusion of cryptographic checksums and authentication fields. FlexRay [3], another alternative, provides fault tolerance, deterministic timing, and improved electromagnetic interference resistance, making it suitable for safety-critical applications. However, widespread adoption of these protocols is limited due to cost, compatibility issues, and increased system complexity. As a result, practical IDS solutions that operate over legacy CAN systems remain highly relevant for the near-term deployment in autonomous vehicles. First, CAN does not have built-in mechanisms for authentication, and any malicious node connected to it can send messages (susceptible to spoofing attacks) [5]. Second, CAN is a broadcast network, and all nodes connected to it receive every message. Thus, a malicious node can inject a message to affect the entire network. Third, messages on the CAN bus are not encrypted, thus a malicious node connected to it can see all messages sent by all other nodes.

The commonly known attacks on IVN are Denial of Service (DoS), Distributed DoS (DDoS), fuzzy, spoofing, and replay attacks. DoS attacks are caused by flooding messages on the communication bus, saturating the network and thus blocking normal traffic. DDoS attacks DoS attacks that originate from multiple origins i.e., electronic component units (ECUs), leading to more effective communication disruptions. The AV under a DoS/DDoS attack fails to perform normal functions e.g., brake control, leading to catastrophic incidents. In fuzzy

Received 10 December 2024; revised 10 July 2025; accepted 31 October 2025. Date of publication 12 December 2025; date of current version 26 December 2025. This publication was made possible by TUBITAK-QNRF joint Funding Program (Tubitak-QNRF 4th Cycle grant # ICC04-0812-210017) from the Qatar National Research Fund (a member of Qatar Foundation). The findings herein reflect the work, and are solely the responsibility of the authors. The Associate Editor for this article was A. Amditis. (Corresponding author: Muhammad Asif Khan.)

Muhammad Asif Khan and Hamid Menouar are with Qatar Mobility Innovations Center, Qatar University, Doha, Qatar (e-mail: asifk@ieee.org). Mohamed Abdallah is with Hamad Bin Khalifa University, Doha, Qatar. Digital Object Identifier 10.1109/TITS.2025.3628885

TABLE I
A SUMMARY OF RELATED WORK ON IDS FOR IVN

Ref	Year	Dataset	Model	Performance				
				Ac	Pr	Re	F1	FPR
GIDS [9]	2018	CHD	GAN	97.9	96.8	99.6	-	-
Han et al. [10]	2018	IDS-Challenge, SA	Statistical	-	-	99.9	99.8	-
CANet [11], [12]	2020	CHD	CNN	99.4	-	-	-	-
Khan et al. [13]	2022	CHD	Bi-LSTM	99.1	99.1	98.4	99.0	-
STParNet [12]	2023	CHD	CNN+LSTM	99.9	-	-	-	-
Khan et al. [14]	2018	CHD	Statistical	-99.6	99.8	99.3	99.5	-
Zhang et al. [12], [15]	2023	CHD	CNN (MobileNetV3)	99.9	-	-	-	-
Zhang et al. [12], [16]	2023	CHD	CNN (EfficientNet)	99.9	-	-	-	-
Song et al. [17]	2020	CHD	ANN	-	99.6	99.3	99.4	-
Song et al. [17]	2020	CHD	SVM	-	99.8	98.5	99.1	-
Song et al. [17]	2020	CHD	Inception-ResNet	-	99.9	99.8	99.9	-
Baldini et al. [18]	2023	CHD	CNN	99.2	98.5	98.0	-	-
Mehedi et al. [19]	2021	CHD	LSTM	97.6	98.0	93.9	88.8	-
Mehedi et al. [19]	2021	CHD	CNN	95.9	89.1	78.9	88.5	-
Mehedi et al. [19]	2021	CHD	RNN	96.4	91.8	75.9	90.9	-
Mehedi et al. [19]	2021	CHD	NN	95.6	84.2	85.7	89.0	-
Mehedi et al. [19]	2021	CHD	LeNet (CNN)	98.1	98.1	98.0	97.8	-
Mehedi et al. [19]	2021	CHD	ResNet (CNN)	97.9	89.5	88.4	90.0	-
Mehedi et al. [19]	2021	CHD	Inception (CNN)	98.0	91.5	92.6	90.2	-
Baldini et al. [18]	2023	ROAD	CNN	99.7	97.0	94.0	-	-

attacks, the attacker transmits random or deliberately crafted messages to cause errors or unusual behavior, such as disabling safety-critical systems. Though fuzzy attacks can overwhelm the IVN similar to DoS attacks causing communication disruption, their primary goal is to analyze the behavior of one or more ECUs to fuzzy attacks to craft more sophisticated exploits. In spoofing attacks, the attacker impersonates a legitimate ECU by injecting malicious messages into the network. A spoofing attack typically performs a specific malicious function e.g., triggering an emergency stop, disabling the brakes, etc. In replay attacks, the attacker captures valid IVN messages and replays them later. These can cause out-of-context abnormal behavior e.g., locking the doors or altering speed control. Replay attacks are also called Man-in-the-Middle (MiM) attacks.

There are a couple of recent surveys [1], [3], [6], [7], [8] on intrusion detection in CAVs. A closer look at these surveys and some related works (Table I), highlights an interesting observation i.e., merely all these works report a significantly high detection performance over several metrics. Such an ideal performance can indicate two possibilities; First, the intrusion detection problem in IVN is easy in general, and simple off-the-shelf machine learning models can be used to detect anomalies and attacks on the IVN. Second, the publicly available datasets used in most of these works are smaller in size and lack complexity, and the problem can be difficult and complex in real-world scenarios.

A. Contribution and Organization

The contribution of this work is as follows:

- A critical analysis of current IDS methods in IVNs, revealing their limitations in generalization and real-world feasibility (Section II);
- A novel Dynamic Windowing Intrusion Detection System (DWIDS) that adapts detection granularity in real-time based on observed metrics (Section III);
- Implementation of both single-frame and multi-frame models, including multi-label classification;
- Extensive evaluation on two public datasets with imbalanced attack distributions, demonstrating superior performance across detection metrics (Section IV).

The results of the evaluation are critically discussed and compared against earlier works (Section V). Interesting conclusions are drawn and directions for future work are highlighted in Section VI.

II. RELATED WORK

A. Methods for Developing Ids

Earlier works on IDS used signature-based methods and traditional machine-learning methods [20] to detect malicious frames in IVNs. For instance, [21] use Recursive least squares (RLS) and cumulative sum (CUSUM) algorithms to detect attacks that use fake time stamps. Similarly, [22] uses the historic frame arrival rates using the sliding window technique to detect malicious/fake frames. However, since these methods rely on frame timestamp data, they often fail to classify aperiodic messages correctly.

Recently, more advanced deep learning architectures such as Convolutional Neural Networks (CNNs) [12], [23], Long Short Term Memory (LSTM) [11], [12], [13], [24], [25], and Generative Adversarial Networks (GANs) [9], [26] are used. CNNs work well on images to automatically extract spatial features from data. Since CAN bus data is sequential and single-dimensional (1D), most works applying CNNs on CAN bus data transform the 1D CAN message frames to

two-dimensional (2D) spatial data (frames). LSTM [27] is a type of recurrent neural network (RNN) designed to learn and retain long-term dependencies in sequential data. LSTM was developed to handle the vanishing gradient problem of standard RNNs [11]. LSTM defines a *window* to segment a continuous data stream into fixed-length intervals or sequences and feed it as input to the LSTM model. A sliding window technique is then used to create overlapping sequences, allowing the model to maintain context as it processes the data. The window size is an important parameter for IDS to measure the response time of the model to potential attacks.

Song et al. [17] proposed an IDS system based on deep CNN (DCNN). Since CNN works best on 2D grids (e.g., images), the sequential frames are stacked together to form 2D grids (frames). These frames are then fed to an Inception-ResNet CNN network. The network reports F1 score of higher than 99.9% on car-hacking dataset [28]. Baldini [18] proposed a CNN-based IDS system for IVN. The CAN bus messages are converted into histograms and features are extracted using correlations of histograms and fed to 1D CNN. The performance is evaluated using the standard metrics, reporting accuracy higher than 99% and car-hacking [28] and ROAD [29] datasets. Hanselmann et al. proposed CANet [11] to combine LSTM with autoencoders [30]. CANet uses multiple independent LSTM models for each CAN ID to capture the temporal patterns in CAN bus data to generate outputs that are aggregated and then fed to an autoencoder for final anomaly detection. CANet achieves a high true negative rate (TNR >0.99). However, the use of separate LSTM models for each CAN ID is almost an impractical approach. Although the authors in [11] used only 13 CAN IDs in their evaluation, it will require a huge number of LSTM sub-networks to implement a fully-fledged IDS system to scan all CAN IDs. For instance, in standard CAN (ISO [4]), the CAN ID is an 11-bit identifier allowing for a total of $2^{11} = 2048$ possible CAN IDs. Similarly, in extended CAN [4], the CAN ID is a 29-bit identifier, allowing for a total of $2^{29} = 536,870,912$ possible CAN IDs.

Khan et al. [13] proposed an IDS based on bidirectional-LSTM evaluated on the UNSWNB-15 dataset [31] and car hacking dataset (CHD) [28] reporting accuracy of 98.88% and 99.11%, respectively. They also compared the results against other methods including SVM, NB, DT, and ANN. The Bi-LSTM outperformed all methods. ANN (on CHD) achieved the second-highest recall and F1 score. The work however neither describes the architecture of the LSTM model, nor the window size. Zhang et al. [12] proposed a Convolution-LSTM evaluated on CHD [28] with a reported accuracy of higher than 99% on all attacks. The results are compared against MobileNetV3 [15], EfficientNet [16], and CANet [11] and showed higher accuracy scores than all models. Like [13], the study does not provide details about the window size used in the LSTM model. Reference [32] also proposed an encoder-decoder network (CNN as decoder and LSTM as decoder) for IDS, reporting 99% accuracy and 0.32% false-positive rate on ROAD dataset [29]. Authors in [9] propose a Generative Adversarial Network (GAN) for intrusion detection in CAN bus data. The CAN IDs from CAN frames are used

to extract patterns in CAN IDs and represented as a 2D image (called CAN image) using a one-hot encoder. Authors consider different input sizes in the range of 32-128 CAN IDs to form the CAN image. Since each CAN frame is separately labeled in the original dataset as normal/attack, the CAN image is labeled as “attack” if at least one CAN ID in the CAN image belongs to an attack frame. Then a GAN model is trained on the CAN images to detect intrusions. In [10], authors used the survival analysis method to classify CAN frames as normal or attack. The authors defined chunks of different sizes in the range of 50-300 frames using only the CAN ID of the frame. Recently, [14] proposed an IDS system based on statistical characteristics of CAN frames using a window-based method to classify chunks of CAN frames. Window sizes in the range of 50-1000 are used, whereas CAN ID is used as a feature. The method efficiently detects fuzzy, merge, and denial-of-service (DoS) attacks.

A summary of these works is presented in Table I. Different than the above methods, [23] uses CNN to identify the CAN message source using the physical characteristics of the CAN bus channel. Reference [26] used GANs to detect bus-off, spoofing, masquerade, replay, fuzzy, and same origin method execution (SOME) attacks. Reference [33] proposed authentication between electronic control units to secure the CAN bus. Reference [34] used anomaly detection method based on Self-Supervised Learning (SSL) to detect frame-level attacks without using labeled data. A closely related work is [35], in which the authors proposed a two stage IDS for IVN. In the first stage, the model predicts the traffic as normal and attack. In the second stage, the model identifies the type of attack. Some recent works have also used large language models (LLMs) such as BERT [36], but these models are not suitable for real-time performance due to the high computational loads.

While many recent IDS studies report high accuracy (often $>99\%$), most rely on narrow evaluation settings e.g., single dataset (often balanced), static window sizes, or single attack types, and lack analysis of model generalizability or real-time feasibility. For instance, most 2D-CNN-based models [12], [23] incur high computational load due to the deeper architectures used in them and are not lightweight enough for embedded deployment. LSTM-based architectures [12], [13] show strong temporal modeling but scale poorly when applied per CAN ID. GAN-based methods [26] are effective in detecting unseen patterns but lack interpretability and exhibit unstable training. LLM based methods [36], [37], [38] incur huge computational load and are not feasible for real-time inference. In contrast, our DWIDS is model-agnostic, supports multi-label detection, and uses adaptive windowing, making it more suitable for real-world latency-sensitive deployment. Unlike most earlier works that used a single dataset or a single vehicle model (Table I), this paper validates the performance over cross-vehicle datasets.

B. Analysis

The review of the aforementioned works notes some important observations;

1) *Choosing Features*: Some earlier studies used timestamps to derive frames' inter-arrival times, whereas most

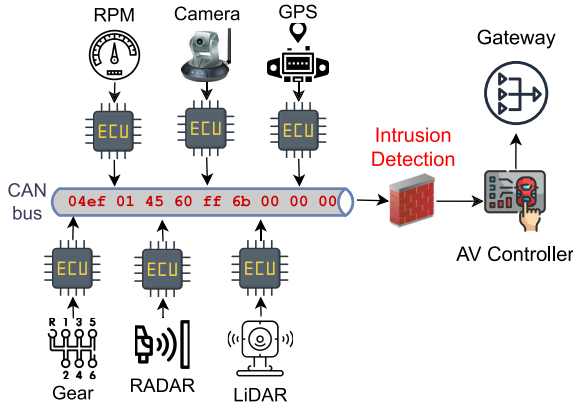


Fig. 1. Intrusion detection in IVN: The ECUs transmit unencrypted sensor data frames over the CAN bus. The IDS monitors the CAN bus transmissions to detect malicious frames before they trigger an action at the AV controller.

recent studies use only CAN ID as a feature. The reason to use only timestamps in earlier studies is that these studies are using signature techniques e.g., the inter-arrival time of frames are compared against historical frames to detect abnormalities. The motivation to use CAN IDs is that they are attributes of different ECUs and reflect the transmission patterns of individual ECUs in a chunk of frames. The payload (data) field is often not used as a feature because the content of data fields is associated with the individual manufacturers. The exclusion of the payload field is thus an attempt to develop universal models that are free and can be deployed in cross-platform vehicles. However, the payload field contains the most important information that can be used to train a model, which can effectively enhance the model's performance.

2) *Choosing Sample Size*: An important aspect is the chunk size. Some studies use IDS that classify each individual frame. However, many studies use only the CAN ID field as a feature by combining CAN IDs from multiple frames into chunks and label each chunk as normal or attack chunks. This approach is completely disintegrated from the earlier approaches because using CAN ID only as a feature can not be used in single-frame prediction. The size of this chunk (obtained via sliding window) varies across studies. Intuitively, this is a modeling aspect and should be carefully chosen. For example, in a DoS attack, the attacker sends a large number of dummy frames, and thus a longer window may be needed to identify the DoS attack pattern. Spoofing attacks are typically intermittent or involve sporadic messages. A shorter chunk may help detect sudden changes in communication quickly. Replay attacks typically involve repeating specific patterns. The chunk size needs to be long enough to capture enough instances of these patterns. Therefore, *the chunk size should be large enough to capture the normal variation in traffic but short enough to detect anomalies promptly (single frame ideally)*. A shorter window provides quicker detection of anomalies but may increase false positives. It can be more responsive to sudden changes or transient attacks. A longer window offers better context and may reduce false positives by considering more data, but they might delay the detection of fast-moving attacks.

3) *Choosing Evaluation Metrics*: Most of these works report performance evaluation over several classification metrics including accuracy, precision, recall, F1 score, and false positive rate (FPR), without explicitly investigating their importance in the context of the dataset used and the available attack scenario. By definition, some of these metrics are not as useful measures as others. For example, most of the datasets for IVN are imbalanced datasets with a huge number of normal and few attack messages. Thus, achieving a higher accuracy score is merely due to the large number of normal messages and is not an indication of the model detection of attacks. Thus, it is important to use the appropriate metrics for a given attack model. Additionally, some authors e.g., [35] proposed two stage detection. We believe such models will incur huge processing and the inference delay encountered will not be suitable for real-time applications like in autonomous vehicles.

III. PROPOSED SCHEME

We first evaluate the two formal approaches i.e., (i) single frame intrusion detection, and (ii) multi-frame (chunk) intrusion detection, with few modifications to achieve high model generalization, and then present a novel approach that best suit real-world implementations in autonomous vehicles. Fig. 2 depicts the two approaches (single frame and multi-frame) for developing an IDS for IVNs. To implement the IDS, the CAN bus data need the following pre-processing steps to train any ML/DL model.

A. Pre-Processing

In both approaches, the raw frames are first pre-processed to extract the training dataset (or data for inference during production). An ML/DL model is then trained using the train sets. Some common pre-processing steps involved are: In case of separate datasets for each type of attacks, the datasets for each attack are merged. The timestamps are removed. The CAN ID and Data fields are converted to decimal values and then to tensors. The attack label T is first replaced by $[N, D, F, G, P]$ to denote Normal, DoS, Fuzzy, Gear, and RPM spoofing frames. Then the labels are mapped to integer values $[0, 1, 2, 3, 4]$ respectively.

B. Method 1: Per-Frame Detection Analysis

In this analysis, the processed dataset is used directly, such that each frame serves as a single sample. The individual frame labels are used to evaluate the performance over the five metrics. In this analysis, we assumed two scenarios; first when the CAN ID is used as a feature alongside the Data field, and second when the CAN ID is not used as a feature (only the Data field). Although it does not make any sense to drop the CAN ID from the feature space if it plays a role in improving the accuracy because it is always visible in CAN frames, the purpose of dropping the CAN ID in the latter approach is to investigate if there is a strong dependency of the accuracy on the CAN ID or not. In other words, it ensures that the model is not learning or overfitting to a single feature (CAN ID) and would be able to generalize to malicious frames with different CAN IDs.

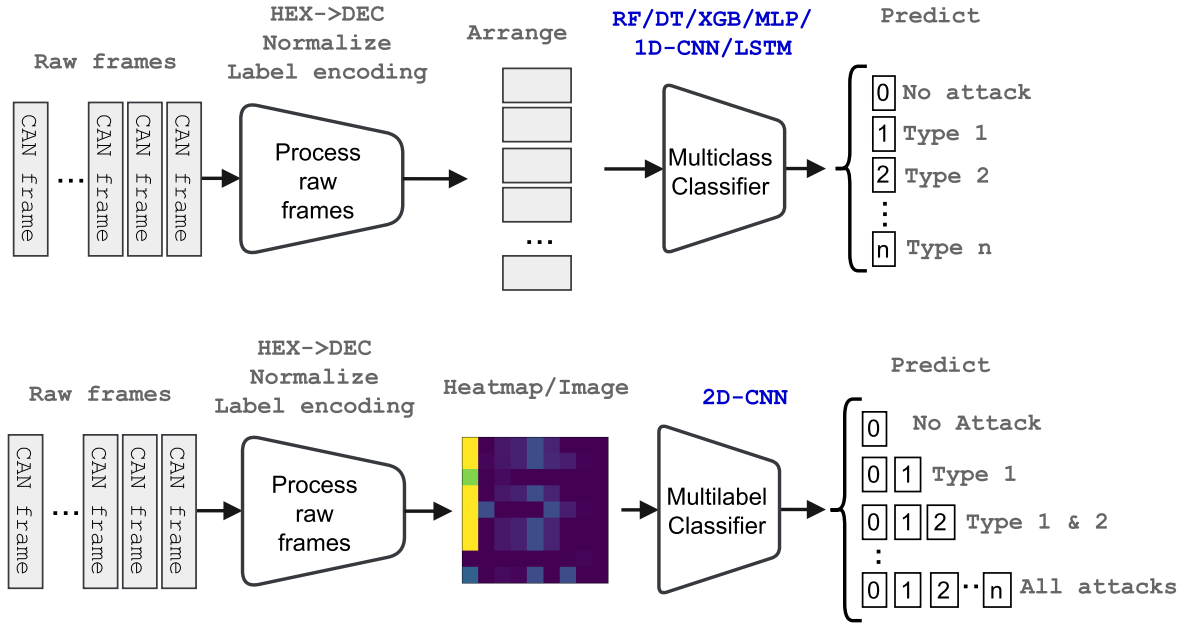


Fig. 2. The proposed scheme showing a 1D multi-class ML/DL model (top) and multi-label classification using 2D-CNN model (bottom) for IDS.

C. Method 2: Frame Sequence Analysis

In this analysis, CAN frames (in the original order) are stacked to form a single sample. The frame stacking is typically done using the timestamp value i.e., Starting with the first frame, the frames are stacked in chunks such that all frames are transmitted during the time window (e.g., 0.5s) and are extracted to form one sample. The process is repeated until it reaches the last chunk.

The frames in each chunk are then arranged to form square 2D images (a single-channel heatmap and a 3-channel RGB image). Since the CAN frames are not transmitted using a fixed frequency, the number of frames in each chunk is different. Thus, padding is applied while creating square images. If the dataset does not have timestamps, fixed-sized chunks are extracted to form squared heatmaps and images, and padding is used in the last chunk. All chunks are then labeled carefully. Since in the original dataset, each frame is independently labeled, it is fairly possible a chunk (sample) will contain more than one type of attack frame in addition to normal frames. The goal is to train all possible attacks in a given chunk (which represents a fixed time window). This process is presented in Algorithm 1.

D. The Proposed Dynamic Windowing Ids (DWIDS) Scheme

We propose a dynamic windowing IDS (DWIDS) method in which the size of the window adapts based on the type of traffic or detected anomalies. This can provide more flexibility and improve detection performance. The proposed method regularly assesses the trade-offs between detection speed and accuracy by monitoring metrics like false positives, false negatives, and detection latency to fine-tune the window size. More specifically, these metrics are used in a feedback loop to fine-tune the window size dynamically, ensuring that the IDS maintains a balance between fast detection (to prevent further attacks) and accurate identification (to reduce false

Algorithm 1 Process CAN Data for Multi-Label Classification

```

1: Input: CAN data (DataFrame), time slot (column), labels (0-4)
2: Output: 2D heatmaps, 3D RGB images, multi-labels
3: Initialize empty lists: heatmaps, images, labels
4: for each chunk in DataFrame do
5:   chunk_data ← Extract rows from DataFrame based on timeslot (0.5s)
6:   X ← chunk_data.iloc[:, 1:10] ▶ Columns 1 to 9
7:   y ← chunk_data.iloc[:, label_column] ▶ Extract labels
8:   if size of X cannot be transformed into square then
9:     padding_size ← Calculate required padding
10:    X ← Pad X with zero-values to make it square
11:   end if
12:   heatmap ← Convert X to 2D heatmap
13:   image ← Convert X to 3D RGB image
14:   Initialize multi_label ← np.zeros(5) ▶ For 5 classes
15:   for each lin np.unique(y) do
16:     multi_label[1] ← 1 ▶ Set index to 1 if class is present
17:   end for
18:   Append heatmap to heatmaps
19:   Append image to images
20:   Append multi_label to labels
21: end for return heatmaps, images, labels

```

alarms). The system relies on real-time analysis and optimization techniques, such as heuristics or machine learning, to make data-driven decisions about window adjustments. The proposed DWIDS is presented in Algorithm 2 and is explained as follows:

Algorithm 2 Dynamic Windowing Intrusion Detection System (DWIDS)

```

1: Initialize window size  $W \leftarrow W_0$  ▶Start
2: Set thresholds:  $FP_{\max}$ ,  $FN_{\max}$ ,  $L_{\max}$  ▶Max acceptable metrics
3: Initialize performance metrics:  $FP$ ,  $FN$ ,  $L \leftarrow 0$ 
4: while IDS is active do
5:   Chunks collected using window  $W$ 
6:   Analyze data and detect anomalies
7:   if Anomaly detected then
8:      $W \leftarrow \max(W_{\min}, W - \Delta W)$  ▶Reduce  $W$ 
9:   else
10:     $W \leftarrow \min(W_{\max}, W + \Delta W)$  ▶Expand  $W$ 
11:   end if
12:   Update metrics:  $FP$ ,  $FN$ ,  $L$ .
13:   if  $FP > FP_{\max}$  or  $FN > FN_{\max}$  or  $L > L_{\max}$  then
14:     Fine-tune  $W \leftarrow W \pm \Delta W$  ▶Adjust  $W$ 
15:   end if
16:   Wait for the next monitoring interval
17: end while

```

The window size W is initialized with W_0 . Environment-specific thresholds are defined for false positives (FP), false negatives (FN), and latency (L). The IDS collects traffic data chunks within the current window (W) and performs anomaly detection. If an anomaly is detected, the window size is reduced to allow more precise monitoring. If no anomaly is detected, the window size is expanded to enhance efficiency. At the end of each detection, the performance metrics (FP, FN, and L) are updated. If these metrics exceed predefined thresholds, the window size is fine-tuned further to balance accuracy and speed. The system waits for the next monitoring interval and continues the process.

To ensure real-time applicability in AV environments, DWIDS was designed to be model-agnostic and compatible with lightweight architectures. Our experiments included efficient implementations such as 1D-CNN and MLP, which achieved high detection performance with low latency. These models can be deployed on embedded automotive platforms with constrained resources, supporting the practical feasibility of DWIDS without requiring heavy computational loads. The DWIDS framework is designed with deployment feasibility in mind. Compared to state-of-the-art models that require per-CAN ID LSTM instances or large CNN backbones, DWIDS supports compact classifiers (e.g., 1D-CNN, MLP) that can execute within the memory and latency constraints of automotive-grade ECUs (e.g., 10–50 ms inference time on NVIDIA Jetson Nano). Moreover, the dynamic windowing mechanism can be tuned for different latency-accuracy trade-offs depending on application criticality (e.g., braking vs. infotainment subsystems).

IV. EXPERIMENTS AND EVALUATION

A. Datasets

We used two datasets namely HCRL car-hacking (CHD) [28] and intrusion detection challenge (IVN-IDS) [10] in this study. The selection of the CHD [28] and IVN-IDS [10]

TABLE II
CH DATASET [28] STATISTICS

	Frame Type	Count	Unique CAN IDs
DoS	Normal (R)	587521	25
	Attack (T)	3047062	1
Fuzzy	Normal (R)	491847	35
	Attack (T)	3259177	2048
Gear Spoofing	Normal (R)	654897	25
	Attack (T)	3925329	1
RPM Spoofing	Normal (R)	597252	25
	Attack (T)	3805725	1

datasets is motivated by their widespread use in recent IDS research, their availability in standard CAN frame format, and the inclusion of multiple attack types (e.g., DoS, fuzzy, spoofing, and replay). These characteristics allow for reproducible benchmarking and compatibility with both single-frame and chunk-based detection models. However, the datasets also have few limitations. First, they are collected from a limited set of vehicles and may not generalize to all vehicle types or brands. Second, the attacks use synthetic or replayed attack traffic that may not fully reflect adversarial behavior in real-world CAN bus scenarios. Third, environmental and driving context data (e.g., speed, weather, driver state) are typically absent, which limits holistic behavioral analysis.

1) *HCRL Car-Hacking [28]*: The car hacking (CH) dataset is constructed from CAN traffic collected via the OBD-II port from a real vehicle. Four different attack messages are injected including DoS attacks, fuzzy attacks, gear spoofing attacks, and RPM gauge spoofing attacks. The dataset contains 300 intrusions, each 3 to 5 seconds long. In DoS attacks, a message with CAN ID: '0000' is injected every 0.3 milliseconds. In fuzzy attacks, messages with random CAN ID are injected every 0.5 milliseconds. In spoofing attacks, messages with specific CAN ID related to RPM/gear information are injected every 1 millisecond. The dataset is publicly available as comma-separated value (CSV) files. Each CSV file contains CAN messages in a specific format given as follows:

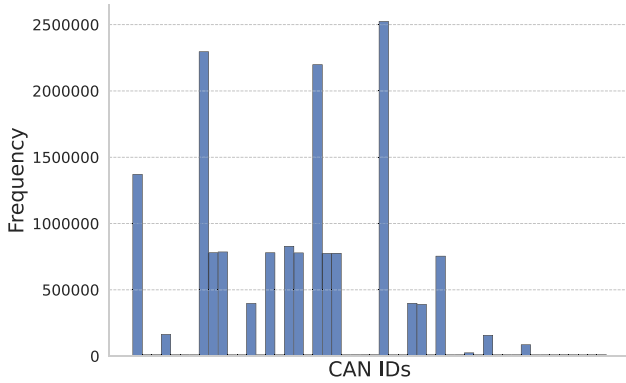
2) *HCRL Intrusion Detection Challenge*: The Intrusion Detection Challenge dataset [10] is a rich dataset of CAN bus data collected from HYUNDAI Sonata, KIA Soul, and CHEVROLET Spark. The dataset contains three types of attacks including Fuzzy, Malfunction, and Replay. The dataset contains (i) timestamp, (ii) CAN ID, (iii) Data length code, (iv) Data, and (v) Label. The fuzzy attacks consist of randomly generated CAN messages. Both the CAN IDs and Data fields are randomly generated. The CAN IDs in attack frames ranged from 0×000 to $0 \times 7FF$ and contained both actual CAN IDs of the vehicle and non-existing CAN IDs (which do not belong to the same vehicle). The malfunction attacks used selected CAN ID from the same vehicle, with a simulated Data field. Replay attacks involved injecting past normal data from the same vehicle into the CAN bus.

The dataset is further analyzed in Table III.

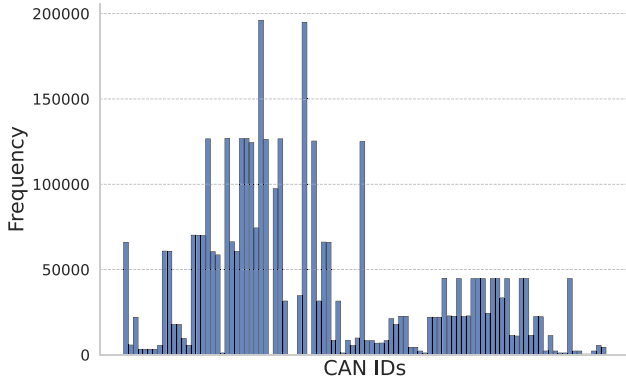
There is no Flag field in the dataset to identify attack messages. However, messages with CAN ID: '0000' indicate a DoS attack, and CAN ID: '0x164' indicated impersonation

TABLE III
IVN IDS CHALLENGE DATASETS [10] STATISTICS

	Frame Type	Count	Unique CAN IDs
Normal	Normal (R)	904347	50
	Attack (T)	0	0
Fuzzy	Normal (R)	659104	50
	Attack (T)	204195	50
Malfunction	Normal (R)	1234479	50
	Attack (T)	138548	2
Replay	Normal (R)	552534	50
	Attack (T)	274	2



(a) car-hacking



(b) IDS Challenge

Fig. 3. Histogram of CAN IDs in car-hacking dataset.

attacks. The traffic with timestamps between 0 sec - 250 sec are all attack-free messages, whereas all traffic after 250 sec contains attack messages mixed with normal traffic. Fig. 3 shows the distribution of CAN IDs in both datasets.

B. Threat Model

In this study, we use the four common attacks (i.e., DoS, Gear Spoofing, RPM Spoofing, Fuzzy) used in several studies and available in the existing datasets. Fig. 4 depicts the raw CAN frames from the car-hacking dataset [28]. It can be observed that the DoS attack is when a bulk of messages are injected with high-priority CAN ID (0000) either at regular or varying intervals. The fuzzy attack is also the injection of bulk frames with random CAN IDs. On the other hand, the spoofing

attacks (both Gear and RPM spoofing) involve the injection of frames with the specific ID of the particular ECU responsible for controlling gear and RPM sensors. Intuitively, the former two attacks (DoS and fuzzy) can be detected by modeling the data in sequences whereas the latter two attacks (gear and RPM) need to be detected by analyzing each (single) frame because even a single spoofing frame can cause the attack to alter the AV response/action.

C. Metrics

The proposed IDS system is evaluated using standard classification metrics i.e., accuracy, precision, recall, and false positive rate (FPR) used throughout the related works. These metrics are calculated as follows 1:

$$\left\{ \begin{array}{l} \text{Accuracy (Ac)} = \frac{TP+TN}{TP+TN+FP+FN} \\ \text{Precision (Pr)} = \frac{TP}{TP+FP} \\ \text{Recall / TPR (Re)} = \frac{TP}{TP+FN} \\ \text{FPR} = \frac{FP}{FP+TN} \\ \text{TNR (1-FPR)} = \frac{TN}{TN+FP} \\ \text{F1 score (Fs)} = 2 \times \frac{Pr \times Re}{Pr+Re} \end{array} \right. \quad (1)$$

where, TP, FP, TN, and FN denote true positive, false positive, true negative, and false negative. Accuracy is the proportion of all samples ($TP+TN+FP+FN$) correctly detected ($TP+TN$). Precision is the proportion of correctly predicted positive samples (TP) among all positive predictions ($TP+FP$). Recall is the proportion of true positive predictions among all positive samples i.e., it measures the proportion of actual positives that are correctly identified. It is also called *sensitivity* or True Positive Rate (TPR). FPR is the proportion of actual negatives ($FP+TN$) that are incorrectly identified as positives (FP). TNR is the proportion of actual negatives that are correctly identified. TNR is also called *specificity*. The F1 score is the harmonic mean of precision and recall. A low F1 score suggests that either precision or recall (or both) are poor. Accuracy can be misleading in imbalanced datasets where one class dominates. Precision focuses on the quality of positive predictions, penalizing false positives (FP). Recall focuses on capturing all actual positives, and penalizing false negatives (FN). High precision means few false positives, whereas high Recall means most positive samples are identified. As you increase precision (fewer false positives), Recall might drop (more false negatives), and vice versa. FPR gives a trade-off between identifying true positives and avoiding false positives. FPR and Recall highlight the balance between detecting positives and avoiding false positives. TNR is complementary to FPR since, $TNR = 1 - FPR$.

In the context of attacks on IVNs, the metrics should be carefully chosen. For example;

- Attack messages are typically rare i.e., the datasets are imbalanced, and thus high accuracy is often due to the large number of normal samples, and not the model's ability to detect attacks.
- Precision is crucial to avoid false alarms, whereas Recall is critical to detect actual attacks. In IDS, missing an actual attack is usually more dangerous than flagging a few non-attacks. Thus, recall is a more critical metric than precision.

CAN ID	DLC	Data0	Data1	Data2	Data3	Data4	Data5	Data6	Label
0000	00	00	00	00	00	00	00	00	T
0130	18	80	00	ff	1b	80	0a	e4	R
0000	00	00	00	00	00	00	00	00	T
0131	00	80	00	00	4c	7f	0a	9a	R
0000	00	00	00	00	00	00	00	00	T
0000	00	00	00	00	00	00	00	00	T
0130	18	80	00	ff	1b	80	0a	e4	R
0000	00	00	00	00	00	00	00	00	T
0131	00	80	00	00	4c	7f	0a	9a	R
0000	00	00	00	00	00	00	00	00	T

(a) DoS attack

CAN ID	DLC	Data0	Data1	Data2	Data3	Data4	Data5	Data6	Label
043f	01	45	60	ff	6b	00	00	00	T
043f	01	45	60	ff	6b	00	00	00	T
043f	01	45	60	ff	6b	00	00	00	T
04f0	00	00	00	80	00	67	d1	13	R
043f	01	45	60	ff	6b	00	00	00	T
0440	ff	00	00	00	ff	c4	08	00	R
0316	05	21	68	09	21	21	00	70	R
018f	fe	50	00	00	00	3c	00	00	R
0002	00	00	00	00	00	06	0b	70	R

(c) Gear Spoofing attack

CAN ID	DLC	Data0	Data1	Data2	Data3	Data4	Data5	Data6	Label
00df	8c	ab	f2	26	7a	29	1a	0c	T
06ea	25	10	9c	ed	5b	16	2c	18	T
02fd	3f	bd	68	f3	c3	4f	28	d4	T
02b0	ff	7f	00	05	c1	R	nan	nan	nan
0370	00	20	00	00	00	00	00	00	R
0000	db	80	dc	1b	8d	c1	5f	32	T
0140	00	00	00	00	0a	26	21	5f	R
018f	fe	58	00	00	00	41	00	00	R
0260	19	21	22	30	08	8f	6e	38	R
02a0	64	00	99	1d	97	02	bd	00	R

(b) Fuzzy attack

CAN ID	DLC	Data0	Data1	Data2	Data3	Data4	Data5	Data6	Label
0316	45	29	24	ff	29	24	00	ff	T
02c0	15	00	00	00	00	00	00	00	R
05f0	01	00	R	nan	nan	nan	nan	nan	nan
0316	45	29	24	ff	29	24	00	ff	T
0316	45	29	24	ff	29	24	00	ff	T
0316	05	22	68	09	22	20	00	75	R
018f	fe	3b	00	00	00	3c	00	00	R
0260	19	22	22	30	ff	8f	6e	3f	R
02a0	60	00	83	1d	96	02	bd	00	R
0329	dc	b8	7e	14	11	20	00	14	R

(d) RPM Spoofing attack

Fig. 4. Raw CAN frames showing normal frames (white background) and attack frames (red background).

- TNR (how the model detects normal samples) is also a less critical metric. However, FPR (i.e., 1-TNR) can be critical for system trustworthiness. A high FPR (low TNR) means too many normal messages are incorrectly flagged as attacks. The IDS will thus incorrectly detect a DoS attack.

D. Baselines for Single-Frame Intrusion Detection

1) *Decision Trees*: A decision tree (DT) is a tree-like model where each internal node represents a decision on a feature, each branch represents an outcome of the decision, and each leaf node represents a class label or regression value. A DT splits data into branches, each branch represents a decision rule based on feature values, leading to a final prediction at the leaf nodes. DTs are highly interpretable, simple to implement, and computationally inexpensive to train. DTs do not require feature scaling and can handle both numerical and categorical data. However, DTs are prone to Overfitting, especially when using deep trees.

2) *Random Forest*: Random Forest (RF) is an ensemble learning method that combines multiple decision trees to improve predictive performance and control overfitting. It constructs a “forest” of trees by training them on random subsets of the data and features. RF uses bootstrapping (sampling with replacement) to train each tree on a different subset of the data. RF, compared to a DT, is less prone to overfitting. They are also relatively more accurate and robust to noise and outliers. However, they are more complex to implement and computationally intensive (takes much longer to train and predict).

3) *Extreme Gradient Boosting*: Extreme Gradient Boosting (XGBoost) is an ensemble method that builds a strong model by combining the predictions of multiple decision

trees. XGBoost has high speed and accuracy, particularly on structured/tabular data. It can handle missing data and is relatively less prone to overfitting.

4) *Multilayer Perceptron*: Multilayer Perceptrons (MLPs) is a class of artificial neural networks (ANNs) that are fully connected and consist of multiple layers of neurons. MLPs work well on small to medium-sized tabular datasets and can model complex non-linear relationships. However, MLPs are sensitive to hyperparameters, prone to overfitting, and computationally intensive.

5) *Convolution Neural Networks*: Convolution Neural Networks (CNNs) are a class of deep learning models, primarily used for spatial data (e.g., images, video, and audio). CNNs typically require large amounts of data to avoid overfitting. They use parameter sharing to reduce the number of parameters and computational cost.

6) *Log-Short Term Memory Network*: LSTM (Long Short-Term Memory) is a special type of Recurrent Neural Network (RNN) designed to handle long-term dependencies in sequential data (e.g., time series, natural language, speech, and video), where the order of the elements matters. RNNs (and LSTM) use outputs from the previous time step and fed back into the network along with the current input. Vanilla RNNs are the simplest forms of an RNN, where each time step is treated identically, and the hidden state is updated using a fixed activation function (e.g., tanh). LSTMs can process sequences of varying lengths. They are typically slow to train and complex to tune.

V. RESULTS AND DISCUSSION

We first analyzed the number of CAN IDs in both datasets. These are depicted in Fig. 3. The analysis shows that both datasets have a large number of CAN IDs available to allow

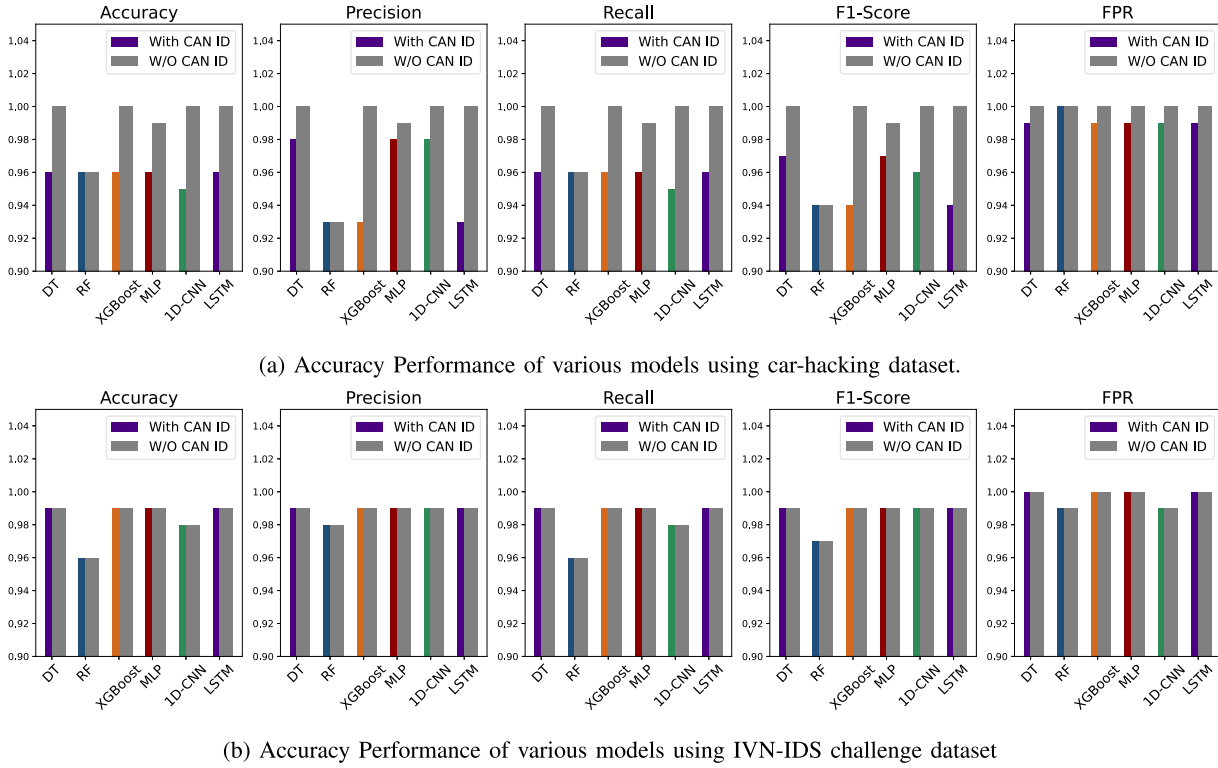


Fig. 5. Performance comparison of six baseline models on the CH dataset [28] for single-frame intrusion detection, showing accuracy, precision, recall, F1-score, and FPR with and without the CAN ID feature.

models to learn from other features and not to overfit to a single feature (CAN ID). Thus, CAN ID can serve as an effective feature in the training of ML/DL models for IDS.

A. Single Frame Attack Detection

We first trained the six baseline models on both datasets separately. The analysis in Table II and III shows that the datasets are imbalanced (large difference in number of each attack and normal frames), thus we used class-weights so that the model is not biased toward the majority class. We used quite simple settings (mostly default settings) for each model, since the focus is not to extensively finetune the models, but to judge the IDS task in general using off-the-shelf models. The decision tree (DT), random forest (RF) are implemented in Scikit-learn, XGBoost is implemented using xgboost library, whereas the MLP, CNN, and LSTM models are implemented using PyTorch (trained up to 10 epochs). The models are trained on fixed size of the train set (70%) and tested on test set (30%) and the results are computed across all metrics. The detailed evaluation is depicted in Fig. 5. The results show that all models fairly achieve very high performance over all metrics. Surprisingly, we repeated the experiments for two separate cases (i) with CAN ID, and (ii) without CAN ID, to understand whether the CAN IDs are responsible for the high accuracy and may lead to overfitting. However, interestingly for both cases, the very high performance scores were achieved for all metrics. Notably, slightly lower performance (still acceptable) are achieved for some methods e.g., RF in few cases.

We further analyzed these results to investigate if the high scores are due to the high number of normal cases, but on the contrary, the models have correctly predicted attack classes too, as depicted in the confusion matrices in Fig. 6 and 7. The confusion matrix shows that in the case of CAN ID is included in the feature set, most models (except DT which was relatively) accurately identifies DoS, gear, and RPM spoofing attacks with close to 100% accuracy. However, a few instances of fuzzy attacks were wrongly predicted. On the other hand, in the case when CAN ID is not included in the feature set, the number of misclassified fuzzy attacks further increased. It shows that the CAN ID plays a role in the detection. The gear and RPM spoofing attacks use the same CAN ID throughout the dataset. Thus are more accurately detected. Fuzzy attacks use different CAN IDs (a total of 2048 CAN IDs are used across fuzzy attacks and the CAN IDs are also used in normal frames) and the data fields contain different values even for the same CAN ID in different fuzzy attack frames. The detection of fuzzy attacks is thus non-trivial and more relatively challenging.

B. Multi-Frame Attack Detection

In this section, the proposed DWIDS model is evaluated over the car-hacking dataset and IVN-IDS challenge datasets, and results are reported in Table IV and V, respectively.

The model achieved high performance across all classes on both datasets, with average precision, recall, and F1 scores in the 0.96-0.98 range. The model also performed consistently well for both majority (“Normal,” 72% in car-hacking and

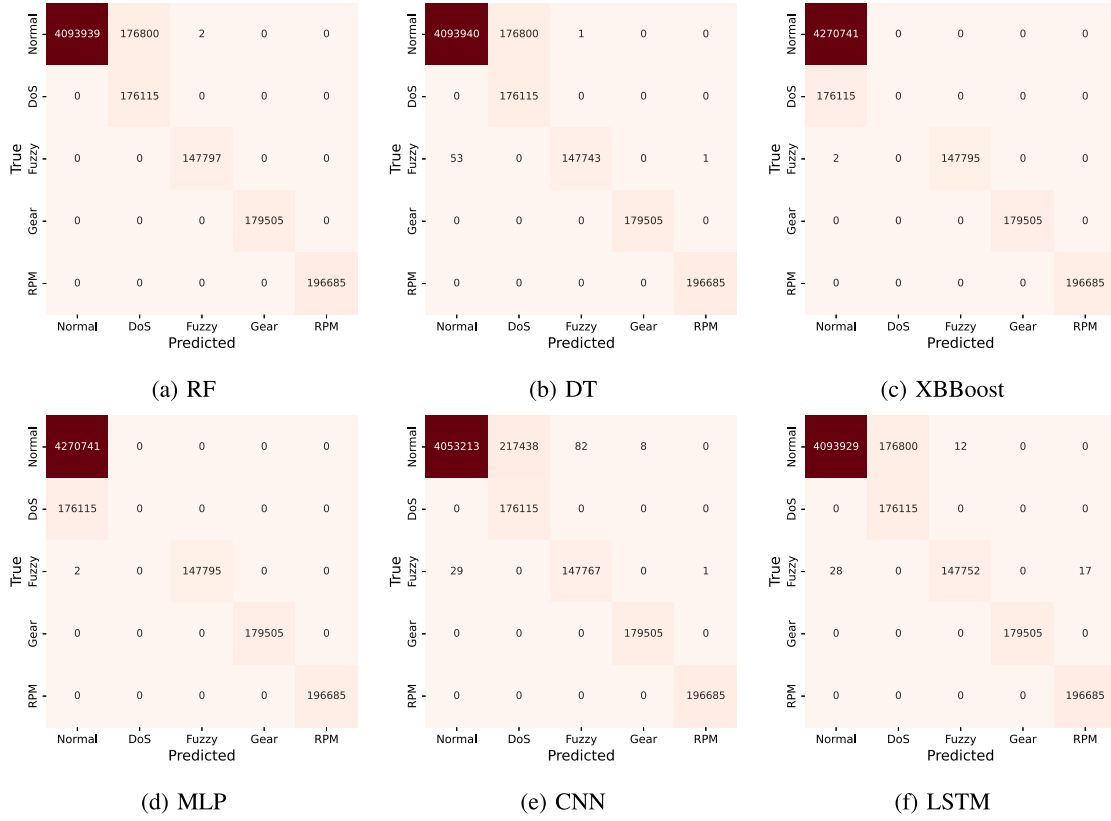


Fig. 6. Confusion Matrix using single-frame intrusion detection (without including CAN ID used as a feature).

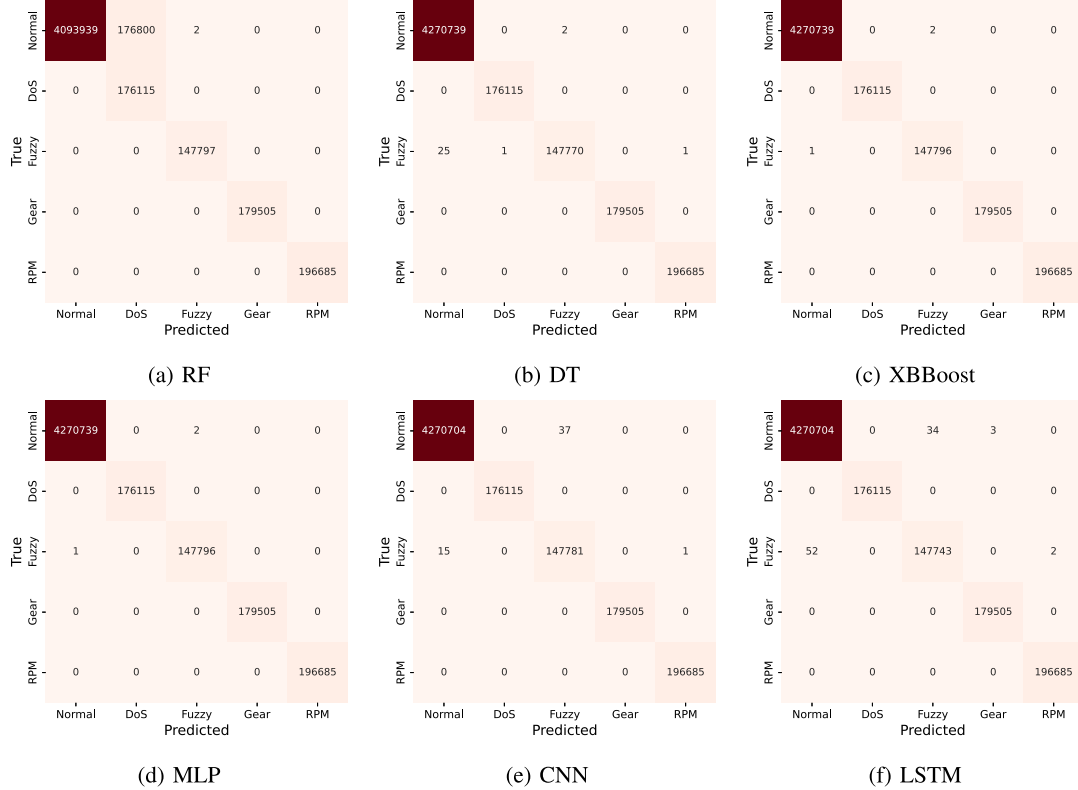


Fig. 7. Confusion Matrix using single-frame intrusion detection (with CAN ID included as a feature).

78% in IVN-IDS challenge) and minority classes (e.g., “DoS” and “Fuzzy”, “Replay”), demonstrating robustness against class imbalance. The weighted average scores in both tables further highlight the model’s effectiveness in handling the

TABLE IV
MULTILABEL CLASSIFICATION USING DWIDS ON
CAR-HACKING DATASET [28]

Class Label	Precision	Recall	F1-score	Support
Normal	0.98	0.97	0.97	72%
DoS	0.97	0.96	0.96	5%
Fuzzy	0.97	0.96	0.95	6%
Gear	0.98	0.97	0.97	8%
RPM	0.98	0.97	0.97	9%
Macro Average	0.98	0.97	0.96	—
Weighted Average	0.98	0.97	0.97	—

TABLE V
MULTILABEL CLASSIFICATION METRICS (IVN-IDS
CHALLENGE DATASET [10])

Class Label	Precision	Recall	F1-score	Support
Normal	0.98	0.98	0.98	78%
Fuzzy	0.97	0.96	0.96	11%
Malfunction	0.96	0.95	0.95	10%
Replay	0.98	0.98	0.98	2%
Macro Average	0.97	0.97	0.97	—
Weighted Average	0.98	0.97	0.97	—

dataset's distribution. Unlike traditional IDS models that are limited to single-label classification or fixed window detection, DWIDS enables simultaneous identification of mixed attack types within short temporal windows. This enhances responsiveness and reliability in real-world driving conditions where multiple attack vectors may manifest concurrently. The consistent performance across two imbalanced datasets confirms DWIDS's robustness and generalization capability.

VI. CONCLUSION AND FUTURE WORK

This study proposes DWIDS, a dynamic, adaptive intrusion detection system for IVNs, designed for real-time, multi-attack, and resource-constrained environments. The system outperforms conventional IDS models across key metrics, even in imbalanced and multi-label scenarios. These results suggest that DWIDS is a practical and scalable solution for securing next-generation autonomous vehicles. In future work, several promising directions can be pursued to enhance the practicality and robustness of IDS for IVNs. First, evaluating DWIDS on datasets with rare or stealthy attack types, such as low-rate spoofing or gradual replay attacks to assess robustness in edge cases. Second, integrating temporal context from external traffic data (e.g., GPS, driver behavior) may improve attack attribution and reduce false positives. Third, lightweight versions of DWIDS should be developed for deployment on embedded ECUs with limited computational resources. Fourth, adversarial robustness should be examined to ensure the IDS can defend against crafted attack inputs. Lastly, combining the DWIDS with in-vehicle authentication and encryption schemes

can provide a layered defense approach, balancing anomaly detection with message integrity.

REFERENCES

- [1] W. Wu et al., "A survey of intrusion detection for in-vehicle networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 3, pp. 919–933, Mar. 2020.
- [2] M. Han, A. Wan, F. Zhang, and S. Ma, "An attribute-isolated secure communication architecture for intelligent connected vehicles," *IEEE Trans. Intell. Vehicles*, vol. 5, no. 4, pp. 545–555, Dec. 2020.
- [3] M. Almeshdhar et al., "Deep learning in the fast lane: A survey on advanced intrusion detection systems for intelligent vehicle networks," *IEEE Open J. Veh. Technol.*, vol. 5, pp. 869–906, 2024.
- [4] Road Vehicles—Controller Area Network (CAN)—Part 1: Data Link Layer and Physical Signalling, Standard 11898-1:2024, 2015.
- [5] C. Shang, J. Cao, J. Liu, Y. Zhang, B. Niu, and H. Li, "CEAMP: A cross-domain entity authentication and message protection framework for intra-vehicle network," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 6780–6795, Jul. 2024.
- [6] B. Lampe and W. Meng, "A survey of deep learning-based intrusion detection in automotive applications," *Expert Syst. Appl.*, vol. 221, Jul. 2023, Art. no. 119771.
- [7] O. Ajibuwa, B. Hamdaoui, and A. A. Yavuz, "A survey on AI/ML-driven intrusion and misbehavior detection in networked autonomous systems: Techniques, challenges and opportunities," 2023, *arXiv:2305.05040*.
- [8] S. Rajapaksha, H. Kalutara, M. O. Al-Kadri, A. Petrovski, G. Madzudzo, and M. Cheah, "AI-based intrusion detection systems for in-vehicle networks: A survey," *ACM Comput. Surv.*, vol. 55, no. 11, pp. 1–40, Feb. 2023.
- [9] E. Seo, H. M. Song, and H. K. Kim, "GIDS: GAN based intrusion detection system for in-vehicle network," in *Proc. 16th Annu. Conf. Privacy, Secur. Trust (PST)*, Aug. 2018, pp. 1–6.
- [10] M. L. Han, B. I. Kwak, and H. K. Kim, "Anomaly intrusion detection method for vehicular networks based on survival analysis," *Veh. Commun.*, vol. 14, pp. 52–63, Oct. 2018.
- [11] M. Hanselmann, T. Strauss, K. Dormann, and H. Ulmer, "CANet: An unsupervised intrusion detection system for high dimensional CAN bus data," *IEEE Access*, vol. 8, pp. 58194–58205, 2020.
- [12] A. Zhang, K. Wang, B. Wang, and Y. Wu, "LiPar: A lightweight parallel learning model for practical in-vehicle network intrusion detection," 2023, *arXiv:2311.08000*.
- [13] I. A. Khan, N. Moustafa, D. Pi, W. Haider, B. Li, and A. Jolfaei, "An enhanced multi-stage deep learning framework for detecting malicious activities from autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 25469–25478, Dec. 2022.
- [14] J. Khan, D.-W. Lim, and Y.-S. Kim, "Intrusion detection system CAN-bus in-vehicle networks based on the statistical characteristics of attacks," *Sensors*, vol. 23, no. 7, p. 3554, Mar. 2023.
- [15] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, CA, CA, USA, Oct. 2019, pp. 1314–1324.
- [16] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.
- [17] H. M. Song, J. Woo, and H. K. Kim, "In-vehicle network intrusion detection using deep convolutional neural network," *Veh. Commun.*, vol. 21, Jan. 2020, Art. no. 100198.
- [18] G. Baldini, "In-vehicle network intrusion detection system using convolutional neural network and multi-scale histograms," *Information*, vol. 14, no. 11, p. 605, Nov. 2023.
- [19] S. T. Mehedi, A. Anwar, Z. Rahman, and K. Ahmed, "Deep transfer learning based intrusion detection system for electric vehicular networks," *Sensors*, vol. 21, no. 14, p. 4736, Jul. 2021.
- [20] A. S. Bin Shibghatullah, "Mitigating developed persistent threats (APTs) through machine learning-based intrusion detection systems: A comprehensive analysis," *SHIFRA*, vol. 2023, pp. 17–25, Mar. 2023.
- [21] K.-T. Cho and K. G. Shin, "Fingerprinting electronic control units for vehicle intrusion detection," in *Proc. 25th USENIX Secur. Symp. (USENIX Secur.)*, 2016, pp. 911–927.
- [22] A. Taylor, N. Japkowicz, and S. Leblanc, "Frequency-based anomaly detection for the automotive CAN bus," in *Proc. World Congr. Ind. Control Syst. Secur. (WCICSS)*, Dec. 2015, pp. 45–49.
- [23] W. Jeong, S. Han, E. Choi, S. Lee, and J.-W. Choi, "CNN-based adaptive source node identifier for controller area network (CAN)," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13916–13920, Nov. 2020.
- [24] S. Longari, D. H. Nova Valcarcel, M. Zago, M. Carminati, and S. Zanero, "CANnlo: An anomaly detection system based on LSTM autoencoders for controller area network," *IEEE Trans. Netw. Service Manage.*, vol. 18, no. 2, pp. 1913–1924, Jun. 2021.

- [25] J. Ashraf, A. D. Bakhshi, N. Moustafa, H. Khurshid, A. Javed, and A. Beheshti, "Novel deep learning-enabled LSTM autoencoder architecture for discovering anomalous events from intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4507–4518, Jul. 2021.
- [26] J. Qin, Y. Xun, Z. Deng, and J. Liu, "GPIDS: GAN assisted contextual pattern-aware intrusion detection system for IVN," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 12682–12693, Sep. 2024.
- [27] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [28] *Hacking and Countermeasure Research Lab*, car-hacking dataset for the intrusion detection, Korea Univ., Seoul, South Korea, 2021.
- [29] M. E. Verma, M. D. Iannacone, R. A. Bridges, S. C. Hollifield, B. Kay, and F. L. Combs, "Road: The real ornl automotive dynamometer controller area network intrusion detection dataset (with a comprehensive can IDS dataset survey & guide)," 2020, *arXiv:2012.14600*.
- [30] P. Baldi, "Autoencoders, unsupervised learning, and deep architectures," in *Proc. ICML Workshop Unsupervised Transf. Learn.*, 2012, pp. 37–49.
- [31] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6.
- [32] H. Alqahtani and G. Kumar, "A deep learning-based intrusion detection system for in-vehicle networks," *Comput. Electr. Eng.*, vol. 104, Dec. 2022, Art. no. 108447.
- [33] J. Cui et al., "Lightweight encryption and authentication for controller area network of autonomous vehicles," *IEEE Trans. Veh. Technol.*, vol. 72, no. 11, pp. 14756–14770, Nov. 2023.
- [34] J. Cao et al., "Anomaly detection for in-vehicle network using self-supervised learning with vehicle-cloud collaboration update," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 7454–7466, Jul. 2024.
- [35] S. Almutlaq, A. Derhab, M. M. Hassan, and K. Kaur, "Two-stage intrusion detection system in intelligent transportation systems using rule extraction methods from deep neural networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 15687–15701, Dec. 2023.
- [36] M. Fu, P. Wang, M. Liu, Z. Zhang, and X. Zhou, "IoV-BERT-IDS: Hybrid network intrusion detection system in IoV using large language models," *IEEE Trans. Veh. Technol.*, vol. 74, no. 2, pp. 1909–1921, Feb. 2025.
- [37] S. Yang et al., "Large language models for network intrusion detection systems: Foundations, implementations, and future directions," 2025, *arXiv:2507.04752*.
- [38] E. Nwafor and H. Olufowobi, "CANBERT: A language-based intrusion detection model for in-vehicle networks," in *Proc. 21st IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2022, pp. 294–299.



Muhammad Asif Khan (Senior Member, IEEE) received the B.Sc. degree in telecommunication engineering from the University of Engineering and Technology, Peshawar, Pakistan, in 2009, the M.Sc. degree in telecommunication engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2013, and the Ph.D. degree in electrical engineering from Qatar University in 2020. He is currently a Research Scientist at Qatar Mobility Innovations Center (QMIC), Doha, Qatar. He published more than 50 peer-reviewed journal articles, conference papers, and book chapters. He also served as a TPC member of numerous conferences, including NeurIPS, ICCV, ICC, Globecom, CCNC, VTC, ICCE, BigData, Sensors, and GEM. He was awarded the Post-Doctoral Research Award (PDRA) from Qatar National Research Fund (QNRF) for his research work as a Post-Doctoral Research Fellow at Qatar University, Doha. He served as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, IEEE TRANSACTIONS ON TECHNOLOGY AND SOCIETY, and IEEE SIGNAL PROCESSING LETTERS.



Hamid Menouar (Senior Member, IEEE) received the Ph.D. degree from the Esteemed French School Telecom Paris Tech in 2008. He is the Principal Research and Development and the Innovation Lead of Qatar Mobility Innovations Center (QMIC). In his role, he spearheads the design and development of cutting-edge solutions aimed at enhancing the quality of life. His leveraging state-of-the-art technologies like AI and the IoT, he focuses on creating smart, efficient, and sustainable cities. He possesses extensive expertise in AI, the IoT, connected and automated mobility, and uncrewed aerial vehicles (UAV) or drones. His research endeavors have yielded a remarkable collection of scientific research publications, both in esteemed international conferences and journals, along with multiple patents. Several of his innovative solutions have played a pivotal role in the success of Qatar FIFA World Cup 2022.



Mohamed Abdallah (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees from the University of Maryland, College Park, MD, USA, in 2001 and 2006, respectively. He is currently a Professor and the Associate Dean of undergraduate studies and quality assurance with the College of Science and Engineering, Hamad Bin Khalifa University, Ar-Rayyan, Qatar. He has authored or co-authored more than 220 articles, contributed to four book chapters, and holds four patents. His research interests include AI for communications, 6G networks, wireless security, electric vehicles, and smart grids. His recognitions include a Research Fellow Excellence Award at Texas A&M University at Qatar in 2016 and multiple IEEE Best Paper Awards. He was also a Nortel Networks Industrial Fellow from 1999 to 2003.